CHAPTER J

DESIGN OF CONNECTIONS

This chapter addresses connecting elements, connectors, and the affected elements of connected members not subjected to fatigue loads.

The chapter is organized as follows:

User Note: For cases not included in this chapter, the following sections apply:
- Chapter K Additional Requirements for HSS and Box-Section Connections
- Appendix 3 Fatigue

J1. GENERAL PROVISIONS

1. Design Basis

The design strength, ϕR_n, and the allowable strength, R_n/Ω, of connections shall be determined in accordance with the provisions of this chapter and the provisions of Chapter B.

The required strength of the connections shall be determined by structural analysis for the specified design loads, consistent with the type of construction specified, or shall be a proportion of the required strength of the connected members when so specified herein.

Where the gravity axes of intersecting axially loaded members do not intersect at one point, the effects of eccentricity shall be considered.

2. Simple Connections

Simple connections of beams, girders, and trusses shall be designed as flexible and are permitted to be proportioned for the reaction shears only, except as otherwise indicated in the design documents. Flexible beam connections shall accommodate end rotations of simple beams. Some inelastic but self-limiting deformation in the connection is permitted to accommodate the end rotation of a simple beam.

3. Moment Connections

End connections of restrained beams, girders, and trusses shall be designed for the combined effect of forces resulting from moment and shear induced by the rigidity of the connections. Response criteria for moment connections are provided in Section B3.4b.

> **User Note:** See Chapter C and Appendix 7 for analysis requirements to establish the required strength for the design of connections.

4. Compression Members with Bearing Joints

Compression members relying on bearing for load transfer shall meet the following requirements:

(a) For columns bearing on bearing plates or finished to bear at splices, there shall be sufficient connectors to hold all parts in place.

(b) For compression members other than columns finished to bear, the splice material and its connectors shall be arranged to hold all parts in line and their required strength shall be the lesser of

   (1) An axial tensile force equal to 50% of the required compressive strength of the member; or

   (2) The moment and shear resulting from a transverse load equal to 2% of the required compressive strength of the member. The transverse load shall be applied at the location of the splice exclusive of other loads that act on the member. The member shall be taken as pinned for the determination of the shears and moments at the splice.

> **User Note:** All compression joints should also be proportioned to resist any tension developed by the load combinations stipulated in Section B2.

5. Splices in Heavy Sections

When tensile forces due to applied tension or flexure are to be transmitted through splices in heavy shapes, as defined in Sections A3.1d and A3.1e, by complete-joint-penetration (CJP) groove welds, the following provisions apply: (a) material notch-toughness requirements as given in Sections A3.1d and A3.1e; (b) weld access hole details as given in Section J1.6; (c) filler metal requirements as given in Section J2.6; and (d) thermal cut surface preparation and inspection requirements as given in Section M2.2. The foregoing provision is not applicable to splices of elements of built-up shapes that are welded prior to assembling the shape.

> **User Note:** CJP groove welded splices of heavy sections can exhibit detrimental effects of weld shrinkage. Members that are sized for compression that are also subjected to tensile forces may be less susceptible to damage from shrinkage if they are spliced using partial-joint-penetration (PJP) groove welds on the flanges and fillet-welded web plates, or using bolts for some or all of the splice.

6. Weld Access Holes

Weld access holes shall meet the following requirements:

(a) All weld access holes required to facilitate welding operations shall be detailed to provide room for weld backing as needed.

(b) The access hole shall have a length from the toe of the weld preparation not less than 1½ times the thickness of the material in which the hole is made, nor less than 1½ in. (38 mm).

(c) The access hole shall have a height not less than the thickness of the material with the access hole, nor less than ¾ in. (19 mm), nor does it need to exceed 2 in. (50 mm).

(d) For sections that are rolled or welded prior to cutting, the edge of the web shall be sloped or curved from the surface of the flange to the reentrant surface of the access hole.

(e) In hot-rolled shapes, and built-up shapes with CJP groove welds that join the web-to-flange, weld access holes shall be free of notches and sharp reentrant corners.

(f) No arc of the weld access hole shall have a radius less than ⅜ in. (10 mm).

(g) In built-up shapes with fillet or partial-joint-penetration (PJP) groove welds that join the web-to-flange, weld access holes shall be free of notches and sharp reentrant corners.

(h) The access hole is permitted to terminate perpendicular to the flange, providing the weld is terminated at least a distance equal to the weld size away from the access hole.

(i) For heavy shapes, as defined in Sections A3.1d and A3.1e, the thermally cut surfaces of weld access holes shall be ground to bright metal.

(j) If the curved transition portion of weld access holes is formed by predrilled or sawed holes, that portion of the access hole need not be ground.

7. Placement of Welds and Bolts

Groups of welds or bolts at the ends of any member that transmit axial force into that member shall be sized so that the center of gravity of the group coincides with the center of gravity of the member, unless provision is made for the eccentricity. The foregoing provision is not applicable to end connections of single-angle, double-angle, and similar members.

8. Bolts in Combination with Welds

Bolts shall not be considered as sharing the load in combination with welds, except in the design of shear connections on a common faying surface where strain compatibility between the bolts and welds is considered.

It is permitted to determine the available strength, ϕR_n and R_n/Ω, as applicable, of a joint combining the strengths of high-strength bolts and longitudinal fillet welds as the sum of (1) the nominal slip resistance, R_n, for bolts as defined in Equation J3-4 according to the requirements of a slip-critical connection and (2) the nominal weld strength, R_n, as defined in Section J2.4, when the following apply:

(a) $\phi = 0.75$ (LRFD); $\Omega = 2.00$ (ASD) for the combined joint.

(b) When the high-strength bolts are pretensioned according to the requirements of Table J3.1 or Table J3.1M using the turn-of-nut or combined method, the longitudinal fillet welds shall have an available strength of not less than 50% of the required strength of the connection.

(c) When the high-strength bolts are pretensioned according to the requirements of Table J3.1 or Table J3.1M using any method other than the turn-of-nut method, the longitudinal fillet welds shall have an available strength of not less than 70% of the required strength of the connection.

(d) The high-strength bolts shall have an available strength of not less than 33% of the required strength of the connection.

In joints with combined bolts and longitudinal welds, the strength of the connection need not be taken as less than either the strength of the bolts alone or the strength of the welds alone.

9. Welded Alterations to Structures with Existing Rivets or Bolts

In making welded alterations to structures, existing rivets and high-strength bolts in standard or short-slotted holes transverse to the direction of load, and tightened to the requirements of slip-critical connections are permitted to be utilized for resisting loads present at the time of alteration, and the welding need only provide the additional required strength. The weld available strength shall provide the additional required strength, but not less than 25% of the required strength of the connection.

User Note: The provisions of this section are generally recommended for alteration in building designs or for field corrections. Use of the combined strength of bolts and welds on a common faying surface is not recommended for new design.

10. High-Strength Bolts in Combination with Existing Rivets

In connections designed as slip-critical connections in accordance with the provisions of Section J3, high-strength bolts are permitted to be considered as sharing the load with existing rivets.

J2. WELDS AND WELDED JOINTS

Welding shall conform to the provisions of the *Structural Welding Code—Steel* (AWS D1.1/D1.1M), hereafter referred to as AWS D1.1/D1.1M, except where those provisions differ from this Specification. This Specification governs where provisions differ from AWS D1.1/D1.1M.

User Note: Examples of provisions in AWS D1.1/D1.1M that differ from the Specification provisions are shown in the Commentary.

<table>
<tr><th colspan="4">TABLE J2.1
Effective Throat of
Partial-Joint-Penetration Groove Welds</th></tr>
<tr><th>Welding Process</th><th>Welding Position
F (flat),
H (horizontal),
V (vertical),
OH (overhead)</th><th>Groove Type
(AWS D1.1,
Figure 5.2)</th><th>Effective Throat</th></tr>
<tr><td>Shielded metal arc (SMAW)</td><td rowspan="2">All</td><td rowspan="2">J or U groove

60° V</td><td rowspan="3">depth of groove</td></tr>
<tr><td>Gas metal arc (GMAW)
Flux cored arc (FCAW)</td></tr>
<tr><td>Submerged arc (SAW)</td><td>F</td><td>J or U groove
60° bevel or V</td></tr>
<tr><td>Gas metal arc (GMAW)
Flux cored arc (FCAW)</td><td>F, H</td><td>45° bevel</td><td>depth of groove</td></tr>
<tr><td>Shielded metal arc (SMAW)</td><td>All</td><td rowspan="2">45° bevel</td><td rowspan="2">depth of groove
minus ⅛ in.
(3 mm)</td></tr>
<tr><td>Gas metal arc (GMAW)
Flux cored arc (FCAW)</td><td>V, OH</td></tr>
</table>

1. Groove Welds

1a. Effective Area

The effective area of groove welds shall be taken as the length of the weld times the effective throat.

The effective throat of a CJP groove weld shall be the thickness of the thinner part joined.

When filled flush to the surface, the effective weld throat for a PJP groove weld shall be as given in Table J2.1 and the effective weld throat for a flare groove weld shall be as given in Table J2.2. The effective throat of a PJP groove weld or flare groove weld filled less than flush shall be as shown in Table J2.1 or Table J2.2, less the greatest perpendicular dimension measured from a line flush to the base metal surface to the weld surface.

User Note: The effective throat of a PJP groove weld is dependent on the process used and the weld position. The design documents should either indicate the effective throat required or the weld strength required, and the fabricator should detail the joint based on the weld process and position to be used to weld the joint.

For PJP groove welds, effective throats larger than those for prequalified PJP groove welds in AWS D1.1/D1.1M, Figure 5.2, and flare groove welds in Table J2.2 are permitted for a given welding procedure specification (WPS), provided the fabricator establishes by testing the consistent production of such larger effective throats. Testing shall consist of sectioning the weld normal to its axis, at mid-length, and at terminal ends. Such sectioning shall be made on a number of combinations of material

TABLE J2.2
Effective Throat of Flare Groove Welds

Welding Process	Flare-Bevel-Groove[a]	Flare-V-Groove
GMAW and FCAW-G	$\frac{5}{8}R$	$\frac{3}{4}R$
SMAW and FCAW-S	$\frac{5}{16}R$	$\frac{5}{8}R$
SAW	$\frac{5}{16}R$	$\frac{1}{2}R$

R = radius of joint surface (is permitted to be assumed equal to $2t$ for HSS, where t is the design wall thickness), in. (mm)
[a]For flare-bevel-groove with $R < \frac{3}{8}$ in. (10 mm), use only reinforcing fillet weld on filled flush joint.

TABLE J2.3
Minimum Effective Throat of Partial-Joint-Penetration Groove Welds

Material Thickness of Thinner Part Joined, in. (mm)	Minimum Effective Throat,[a] in. (mm)
To ¼ (6) inclusive	⅛ (3)
Over ¼ (6) to ½ (13)	3⁄16 (5)
Over ½ (13) to ¾ (19)	¼ (6)
Over ¾ (19) to 1½ (38)	5⁄16 (8)
Over 1½ (38) to 2¼ (57)	⅜ (10)
Over 2¼ (57) to 6 (150)	½ (13)
Over 6 (150)	⅝ (16)

[a]See Table J2.1.

sizes representative of the range to be used in the fabrication. During production of welds with increased effective throats, single pass welds and the root pass of multi-pass welds shall be made using a mechanized, automatic, or robotic process, with no decrease in current or increase in travel speed from that used for testing.

1b. Limitations

The minimum effective throat of a partial-joint-penetration groove weld shall not be less than the size required to transmit calculated forces nor the size shown in Table J2.3. Minimum weld size is determined by the thinner of the two parts joined.

2. Fillet Welds

2a. Effective Area

The effective area of a fillet weld shall be the effective length multiplied by the effective throat. The effective throat of a fillet weld shall be the shortest distance from the root to the face of the diagrammatic weld. An increase in effective throat is permitted if consistent penetration beyond the root of the diagrammatic weld is demonstrated by tests using a given welding procedure specification (WPS), provided the fabricator establishes by testing the consistent production of such larger effective throat. Testing shall consist of sectioning the weld normal to its axis at mid-length and terminal ends. During production, single pass welds and the root pass of multi-pass welds shall be made using a mechanized, automatic, or robotic process with no decrease in current or increase in travel speed from that used for testing.

For fillet welds in holes and slots, the effective length shall be the length of the centerline of the weld along the center of the plane through the throat. In the case of overlapping fillets, the effective area shall not exceed the nominal cross-sectional area of the hole or slot in the plane of the faying surface.

2b. Limitations

Fillet welds shall meet the following limitations:

(a) The minimum size of fillet welds shall be not less than the size required to transmit calculated forces nor the size as shown in Table J2.4. These limitations do not apply to fillet weld reinforcements of groove welds.

(b) The maximum specified size of fillet welds of connected parts shall be

(1) Along edges of material less than ¼ in. (6 mm) thick; not greater than the thickness of the material.

(2) Along edges of material ¼ in. (6 mm) or more in thickness; not greater than the thickness of the material minus 1⁄16 in. (2 mm), unless the weld is especially designated on the design and fabrication documents to be built out to obtain full-throat thickness. In the as-welded condition, the distance between the edge of the base metal and the toe of the weld is permitted to be less than 1⁄16 in. (2 mm), provided the weld size is clearly verifiable.

(c) The minimum length of fillet welds designed on the basis of strength shall be not less than four times the nominal weld size, or else the effective size of the weld shall not be taken to exceed one-quarter of its length.

(d) The effective length of end-loaded fillet welds shall be determined as follows:

(1) For fillet welds with a length up to 100 times the weld size, it is permitted to take the effective length equal to the actual length.

(2) When the length of the fillet weld exceeds 100 times the weld size, the effective length shall be determined by multiplying the actual length by the reduction factor, β, determined as

$$\beta = 1.2 - 0.002\left(l/w\right) \le 1.0 \tag{J2-1}$$

where
l = actual length of end-loaded weld, in. (mm)
w = size of weld leg, in. (mm)

(3) When the length of the weld exceeds 300 times the leg size, w, the effective length shall be taken as $180w$.

User Note: For the effect of longitudinal fillet weld length in end connections upon the effective area of the connected member, see Section D3.

TABLE J2.4
Minimum Size of Fillet Welds

Material Thickness of Thinner Part Joined, in. (mm)	Minimum Size of Fillet Weld,[a] in. (mm)
To ¼ (6) inclusive	⅛ (3)
Over ¼ (6) to ½ (13)	3/16 (5)
Over ½ (13) to ¾ (19)	¼ (6)
Over ¾ (19)	5/16 (8)

[a]Leg dimension of fillet welds. When non-low hydrogen electrodes are used, single pass welds must be used.
Note: See Section J2.2b for maximum size of fillet welds.

(e) Intermittent fillet welds are permitted to be used to transfer calculated stress across a joint or faying surfaces and to join components of built-up members. The length of any segment of intermittent fillet welding shall be not less than four times the weld size, with a minimum of 1½ in. (38 mm).

(f) In lap joints, the minimum amount of lap shall be five times the thickness of the thinner part joined, but not less than 1 in. (25 mm). Lap joints joining plates or bars subjected to axial stress that utilize transverse fillet welds only shall be fillet welded along the end of both lapped parts, except where the deflection of the lapped parts is sufficiently restrained to prevent opening of the joint under maximum loading.

(g) Fillet weld terminations shall be detailed in a manner that does not result in a notch in the base metal subjected to applied tension loads. Components shall not be connected by welds where the weld would prevent the deformation required to provide assumed design conditions.

User Note: Fillet weld terminations should be detailed in a manner that does not result in a notch in the base metal transverse to applied tension loads that can occur as a result of normal fabrication. An accepted practice to avoid notches in base metal is to stop fillet welds short of the edge of the base metal by a length approximately equal to the size of the weld. In most welds, the effect of stopping short can be neglected in strength calculations.

There are two common details where welds are terminated short of the end of the joint to permit relative deformation between the connected parts:

- Welds on the outstanding legs of beam double-angle connections are returned on the top of the outstanding leg and stopped no more than 4 times the weld size and not greater than half the leg width from the outer toe of the angle.
- Fillet welds connecting transverse stiffeners to webs of girders that are ¾ in. (19 mm) thick or less are stopped 4 to 6 times the web thickness from the web toe of the flange-to web fillet weld, except where the end of the stiffener is welded to the flange.

Details of fillet weld terminations may be shown on shop standard details.

(h) Fillet welds in holes or slots are permitted to be used to transmit shear and resist loads perpendicular to the faying surface in lap joints or to prevent the buckling or separation of lapped parts and to join components of built-up members. Such fillet welds are permitted to overlap, subject to the provisions of Section J2. Fillet welds in holes or slots are not to be considered plug or slot welds.

(i) For fillet welds in slots, the ends of the slot shall be semicircular or shall have the corners rounded to a radius of not less than the thickness of the part containing it, except those ends that extend to the edge of the part.

3. Plug and Slot Welds

3a. Effective Area

The effective shear area of plug and slot welds shall be taken as the nominal area of the hole or slot in the plane of the faying surface.

3b. Limitations

Plug or slot welds are permitted to be used to transmit shear in lap joints or to prevent buckling or separation of lapped parts, and to join component parts of built-up members, subject to the following limitations:

(a) The diameter of the holes for a plug weld shall not be less than the thickness of the part containing it plus 5⁄16 in. (8 mm), rounded to the next larger odd 1⁄16 in. (even mm), nor greater than the minimum diameter plus 1⁄8 in. (3 mm) or 2¼ times the thickness of the weld.

(b) The minimum center-to-center spacing of plug welds shall be four times the diameter of the hole.

(c) The length of slot for a slot weld shall not exceed 10 times the thickness of the weld.

(d) The width of the slot shall be not less than the thickness of the part containing it plus 5⁄16 in. (8 mm) rounded to the next larger odd 1⁄16 in. (even mm), nor shall it be larger than 2¼ times the thickness of the weld.

(e) The ends of the slot shall be semicircular or shall have the corners rounded to a radius of not less than the thickness of the part containing it.

(f) The minimum spacing of lines of slot welds in a direction transverse to their length shall be four times the width of the slot.

(g) The minimum center-to-center spacing in a longitudinal direction on any line shall be two times the length of the slot.

(h) The thickness of plug or slot welds in material 5⁄8 in. (16 mm) or less in thickness shall be equal to the thickness of the material. In material over 5⁄8 in. (16 mm) thick, the thickness of the weld shall be at least one-half the thickness of the material, but not less than 5⁄8 in. (16 mm).

4. Strength

(a) The design strength, ϕR_n, and the allowable strength, R_n/Ω, of welded joints shall be the lower value of the base material strength determined according to the limit states of tensile rupture and shear rupture and the weld metal strength determined according to the limit state of rupture as follows.

For the base metal

$$R_n = F_{nBM} A_{BM} \tag{J2-2}$$

For complete-joint-penetration and partial-joint-penetration groove welds, and plug and slot welds

$$R_n = F_{nw} A_{we} \tag{J2-3}$$

For fillet welds

$$R_n = F_{nw} A_{we} k_{ds} \tag{J2-4}$$

where

A_{BM} = area of the base metal, in.^{2} (mm^2)
A_{we} = effective area of the weld, in.^{2} (mm^2)
F_{nBM} = nominal stress of the base metal, ksi (MPa)
F_{nw} = nominal stress of the weld metal, ksi (MPa)
k_{ds} = directional strength increase factor

(1) For fillet welds where strain compatibility of the various weld elements is considered

$$k_{ds} = (1.0 + 0.50\sin^{1.5}\theta) \tag{J2-5}$$

(2) For fillet welds to the ends of rectangular HSS loaded in tension

$$k_{ds} = 1.0$$

(3) For all other conditions

$$k_{ds} = 1.0$$

θ = angle between the line of action of the required force and the weld longitudinal axis, degrees

The values of ϕ, Ω, F_{nBM}, and F_{nw}, and limitations thereon, are given in Table J2.5.

User Note: The base metal check need not be performed for fillet welds as illustrated in the Commentary.

User Note: The instantaneous center method is a valid way to calculate the strength of weld groups consisting of weld elements in various directions that considers strain compatibility of the weld elements.

Strain compatibility is satisfied for a linear weld group with a uniform leg size connecting elements with uniform stiffness that are loaded through the center of gravity and, therefore, the directional strength increase may be used. A linear weld group is one in which all elements are in a line or are parallel.

TABLE J2.5
Available Strength of Welded Joints, ksi (MPa)

<table>
<tr><th>Load Type and Direction Relative to Weld Axis</th><th>Pertinent Metal</th><th>ϕ and Ω</th><th>Nominal Stress, F_{nBM} or F_{nw}, ksi (MPa)</th><th>Effective Area, A_{BM} or A_{we}, in.^{2} (mm^2)</th><th>Required Filler Metal Strength Level[a][b]</th></tr>
<tr><td colspan="6">COMPLETE-JOINT-PENETRATION GROOVE WELDS</td></tr>
<tr><td>Tension—normal to weld axis</td><td colspan="4">Strength of the joint is controlled by the base metal.</td><td>Matching filler metal shall be used. For T- and corner-joints with backing left in place, notch tough filler metal is required. See Section J2.6.</td></tr>
<tr><td>Compression—normal to weld axis</td><td colspan="4">Strength of the joint is controlled by the base metal.</td><td>Filler metal with a strength level equal to or one strength level less than matching filler metal is permitted.</td></tr>
<tr><td>Tension or compression—parallel to weld axis</td><td colspan="4">Tension or compression in parts joined parallel to a weld is permitted to be neglected in design of welds joining the parts.</td><td>Filler metal with a strength level equal to or less than matching filler metal is permitted.</td></tr>
<tr><td>Shear</td><td colspan="4">Strength of the joint is controlled by the base metal.</td><td>Matching filler metal shall be used.[c]</td></tr>
<tr><td colspan="6">PARTIAL-JOINT-PENETRATION GROOVE WELDS INCLUDING FLARE-V-GROOVE AND FLARE-BEVEL-GROOVE WELDS</td></tr>
<tr><td rowspan="2">Tension—normal to weld axis</td><td>Base</td><td>$\phi = 0.75$
$\Omega = 2.00$</td><td>F_u</td><td>See J4</td><td rowspan="9">Filler metal with a strength level equal to or less than matching filler metal is permitted.</td></tr>
<tr><td>Weld</td><td>$\phi = 0.80$
$\Omega = 1.88$</td><td>$0.60F_{EXX}$</td><td>See J2.1a</td></tr>
<tr><td>Compression—connections of members designed to bear as described in Section J1.4(b)</td><td colspan="4">Compressive stress is permitted to be neglected in design of welds joining the parts.</td></tr>
<tr><td rowspan="2">Compression—connections not designed to bear</td><td>Base</td><td>$\phi = 0.90$
$\Omega = 1.67$</td><td>F_y</td><td>See J4</td></tr>
<tr><td>Weld</td><td>$\phi = 0.80$
$\Omega = 1.88$</td><td>$0.90F_{EXX}$</td><td>See J2.1a</td></tr>
<tr><td>Tension or compression—parallel to weld axis</td><td colspan="4">Tension or compression in parts joined parallel to a weld is permitted to be neglected in design of welds joining the parts.</td></tr>
<tr><td rowspan="2">Shear</td><td>Base</td><td colspan="3">Governed by J4</td></tr>
<tr><td>Weld</td><td>$\phi = 0.75$
$\Omega = 2.00$</td><td>$0.60F_{EXX}$</td><td>See J2.1a</td></tr>
</table>

TABLE J2.5 (continued)
Available Strength of Welded Joints, ksi (MPa)

Load Type and Direction Relative to Weld Axis	Pertinent Metal	ϕ and Ω	Nominal Stress, F_{nBM} or F_{nw}, ksi (MPa)	Effective Area, A_{BM} or A_{we}, in.^{2} (mm^2)	Required Filler Metal Strength Level[a][b]
FILLET WELDS INCLUDING FILLETS IN HOLES AND SLOTS AND SKEWED T-JOINTS					
Shear	Base	Governed by J4			Filler metal with a strength level equal to or less than matching filler metal is permitted.
	Weld	$\phi = 0.75$ $\Omega = 2.00$	$0.60F_{EXX}$[d]	See J2.2a	
Tension or compression—parallel to weld axis	Tension or compression in parts joined parallel to a weld is permitted to be neglected in design of welds joining the parts.				
PLUG AND SLOT WELDS					
Shear—parallel to faying surface on the effective area	Base	Governed by J4			Filler metal with a strength level equal to or less than matching filler metal is permitted.
	Weld	$\phi = 0.75$ $\Omega = 2.00$	$0.60F_{EXX}$	See J2.3a	

F_{EXX} = filler metal classification strength, ksi (MPa)
[a]For matching weld metal, see AWS D1.1/D1.1M, clause 5.6.1.
[b]Filler metal with a strength level one strength level greater than matching is permitted.
[c]Filler metals with a strength level less than matching are permitted to be used for CJP groove welds between the webs and flanges of built-up sections transferring shear loads, or in applications where high restraint is a concern. In these applications, the weld joint shall be detailed and the weld shall be designed using the thickness of the material as the effective throat, where $\phi = 0.80$, $\Omega = 1.88$, and $0.60F_{EXX}$ is the nominal strength.
[d]The provisions of Section J2.4(a) are also applicable.

(b) For fillet weld groups concentrically loaded and consisting of elements with a uniform leg size that are oriented both longitudinally and transversely to the direction of applied load, the nominal strength, R_n, of the fillet weld group is permitted to be determined as

$$R_n = 0.85\, F_{nw}A_{wel} + 1.5F_{nw}A_{wet} \tag{J2-6}$$

where

A_{wel} = effective area of longitudinally loaded fillet welds, in.^{2} (mm^2)
A_{wet} = effective area of transversely loaded fillet welds, in.^{2} (mm^2)

User Note: The nominal strength of fillet welds groups consisting of elements that are oriented both longitudinally and transversely to the direction of applied load can also be calculated in accordance with Section J2.4(a) neglecting the directional strength increase.

5. Combination of Welds

If two or more of the general types of welds (groove, fillet, plug, slot) are combined in a single joint, the strength of each shall be separately computed with reference to the axis of the group in order to determine the strength of the combination.

6. Filler Metal Requirements

The choice of filler metal for use with CJP groove welds subjected to tension normal to the effective area shall comply with the requirements for matching filler metals given in AWS D1.1/D1.1M.

User Note: The following User Note Table summarizes the AWS D1.1/D1.1M provisions for matching filler metals. Other restrictions exist. For a complete list of base metals and prequalified matching filler metals, see AWS D1.1/D1.1M, Table 5.3 and Table 5.4.

Base Metal (ASTM)	Matching Filler Metal
A36/A36M ≤ ¾ in. (19 mm) thick	60 and 70 ksi filler metal
A36/A36M > ¾ in. (19 mm) thick, A588/A588M[a], A1011/A1011M, A572/A572M Grade 50 (345) and 55 (380), A913/A913M Grade 50 (345), A992/A992M	SMAW: E7015, E7016, E7018, E7028 Other processes: 70 ksi filler metal
A913/A913M Grade 60 (415) and 65 (450)	80 ksi (550 MPa) filler metal
A913/A913M Grade 70 (485)	90 ksi (620 MPa) filler metal

[a]For corrosion resistance and color similar to the base metal, see AWS D1.1/D1.1M, clause 5.6.2.
Note:
In joints with base metals of different strengths, either a filler metal that matches the higher strength base metal or a filler metal that matches the lower strength and produces a low hydrogen deposit may be used when matching strength is required.

Filler metal with a specified minimum Charpy V-notch toughness of 20 ft-lb (27 J) at 40°F (4°C) or lower shall be used in the following joints:

(a) CJP groove welded T- and corner-joints with steel backing left in place, subjected to tension normal to the effective area, unless the joints are designed using the nominal strength and resistance factor or safety factor, as applicable, for a PJP groove weld

(b) CJP groove welded splices subjected to tension normal to the effective area in heavy shapes, as defined in Sections A3.1d and A3.1e

The manufacturer's Certificate of Conformance shall be sufficient evidence of compliance.

7. Mixed Weld Metal

When Charpy V-notch toughness is specified, the process consumables for all weld metal, tack welds, root pass, and subsequent passes deposited in a joint shall be compatible to ensure notch-tough composite weld metal.

J3. BOLTS, THREADED PARTS, AND BOLTED CONNECTIONS

1. Common Bolts

ASTM A307 bolts are permitted except where pretensioning is specified.

2. High-Strength Bolts

Use of high-strength bolts and bolting components shall conform to the provisions of the RCSC *Specification for Structural Joints Using High-Strength Bolts,* hereafter referred to as the RCSC *Specification*, except where those provisions differ from this Specification. This Specification governs where provisions differ from the RCSC *Specification*.

> **User Note:** Examples of provisions in the RCSC *Specification* that differ from this Specification's provisions are shown in the Commentary.

High-strength bolts in this Specification are grouped according to material strength as follows:

Group 120	ASTM F3125/F3125M Grades A325, A325M, F1852, and ASTM A354 Grade BC
Group 144	ASTM F3148 Grade 144
Group 150	ASTM F3125/F3125M Grades A490, A490M, F2280, and ASTM A354 Grade BD
Group 200	ASTM F3043 and ASTM F3111

Use of Group 144 bolting assemblies shall conform to the provisions of ASTM F3148.

Use of Group 200 high-strength bolting assemblies shall conform to the applicable provisions of their ASTM standard. ASTM F3043 and ASTM F3111 Grade 1 assemblies may be installed only to the snug-tight condition. ASTM F3043 and ASTM F3111 Grade 2 assemblies may be used in snug-tight, pretensioned, and slip-critical connections, using procedures provided in the applicable ASTM standard.

> **User Note:** The use of Group 200 bolting assemblies is limited to specific building locations and noncorrosive environmental conditions by the applicable ASTM standard.

When assembled, all joint surfaces, including those adjacent to the washers, shall be free of scale, except tight mill scale.

(a) Bolting assemblies are permitted to be installed to the snug-tight condition when used in
    (1) Bearing-type connections, except as stipulated in Section E6
    (2) Tension or combined shear and tension applications, for Group 120 bolts only, where loosening or fatigue due to vibration or load fluctuations are not design considerations

(b) Bolts in the following connections shall be pretensioned:

(1) As required by the RCSC *Specification*

(2) Connections subjected to vibratory loads where bolt loosening is a consideration

(3) End connections of built-up members composed of two shapes either interconnected by bolts, or with at least one open side interconnected by perforated cover plates or lacing with tie plates, as required in Section E6.1

(c) The following connections shall be designed as slip-critical:

(1) As required by the RCSC *Specification*

(2) The extended portion of bolted, partial-length cover plates, as required in Section F13.3

The snug-tight condition is defined in the RCSC *Specification*. Bolts to be tightened to a condition other than snug tight shall be clearly identified on the design documents. (See Table J3.1 or J3.1M for minimum bolt pretension for connections designated as pretensioned or slip-critical.)

User Note: There are no specific minimum or maximum tension requirements for snug-tight bolts. Bolts that have been pretensioned are permitted in snug-tight connections unless specifically prohibited on design documents.

When bolt requirements cannot be provided within the RCSC *Specification* limitations because of requirements for lengths exceeding 12 diameters or diameters exceeding 1½ in. (38 mm), bolts or threaded rods conforming to Group 120 or Group 150 materials are permitted to be used in accordance with the provisions for threaded parts in Table J3.2.

When ASTM A354 Grade BC, ASTM A354 Grade BD, or ASTM A449 bolts and threaded rods are used in pretensioned connections, the bolt geometry, including the thread pitch, thread length, head, and nut(s), shall be equal to or (if larger in diameter) proportional to that required by the RCSC *Specification*. Installation shall comply with all applicable requirements of the RCSC *Specification* with modifications as required for the increased diameter and/or length to provide the design pretension.

3. Size and Use of Holes

The following requirements apply for bolted connections:

(a) The nominal dimensions of standard, oversized, short-slotted, and long-slotted holes for bolts are given in Table J3.3 or Table J3.3M.

User Note: Bolt holes with a smaller nominal diameter are permitted. See RCSC *Specification* Table 3.1 for bolt hole fabrication tolerances. See Section J9 for diameters of holes in base plates for anchor rods providing anchorage to concrete.

TABLE J3.1
Minimum Bolt Pretension, kips

Bolt Size, in.	Group 120[a]	Group 144[b] and Group 150[b]	Group 200, Grade 2[c]
½	12	15	–
⅝	19	24	–
¾	28	35	–
⅞	39	49	–
1	51	64	90
1⅛	64	80	113
1¼	81	102	143
1⅜	97	121	–
1½	118	148	–

[a]Equal to 0.70 times the minimum tensile strength of bolts as specified in ASTM F3125/F3125M for Grade A325 rounded off to nearest kip.
[b]Equal to 0.70 times the minimum tensile strength of bolts as specified in ASTM F3125/F3125M for Grade A490 rounded off to nearest kip. Group 144 (F3148) assemblies have the same specified minimum pretension as Group 150.
[c]Equal to 0.70 times the minimum tensile strength of bolts as specified in ASTM F3043 and F3111 for Grade 2, rounded off to the nearest kip.

TABLE J3.1M
Minimum Bolt Pretension, kN

Bolt Size, mm	Group 120[a]	Group 150[b]
M12	49	72
M16	91	114
M20	142	179
M22	176	221
M24	205	257
M27	267	334
M30	326	408
M36	475	595

[a]Equal to 0.70 times the minimum tensile strength of bolts as specified in ASTM F3125/F3125M for Grade A325M bolts, rounded off to nearest kN.
[b]Equal to 0.70 times the minimum tensile strength of bolts as specified in ASTM F3125/F3125M for Grade A490M bolts, rounded off to nearest kN.

(b) Standard holes or short-slotted holes transverse to the direction of the load shall be provided in accordance with the provisions of this Specification, unless oversized holes, short-slotted holes parallel to the load, or long-slotted holes are approved by the engineer of record.

(c) Finger shims up to ¼ in. (6 mm) are permitted in slip-critical connections designed on the basis of standard holes without reducing the nominal shear strength of the fastener to that specified for slotted holes.

User Note: Metric grades manufactured to ASTM F3125/F3125M Grade A325M and A490M are similar to Group 120 (830 MPa) and Group 150 (1 040 MPa), respectively.

TABLE J3.2
Nominal Stress of Fasteners and Threaded Parts, ksi (MPa)

Description of Fasteners	Nominal Tensile Stress, F_{nt}, ksi (MPa)[a][b]	Nominal Shear Stress in Bearing-Type Connections, F_{nv}, ksi (MPa)[c]	
		Threads Not Excluded from Shear Planes—(N)[e]	Threads Excluded from Shear Planes—(X)
A307 bolts	45 (310)	27 (190)[d]	27 (190)[d]
Group 120 (e.g., A325)	90 (620)	54 (370)	68 (470)
Group 144 (e.g., F3148)	108 (750)	65 (450)	81 (560)
Group 150 (e.g., A490)	113 (780)	68 (470)	84 (580)
Group 200 (e.g., F3043)	150 (1 000)	90 (620)[f]	113 (780)[f]
Threaded parts meeting the requirements of Section A3.4	$0.75F_u$	$0.450F_u$	$0.563F_u$

[a] For high-strength bolts subjected to tensile fatigue loading, see Appendix 3.
[b] For nominal tensile strength it is permitted to use the tensile stress area of the threaded rod or bolt multiplied by the specified minimum tensile stress of the rod or bolt material, in lieu of the tabulated values based on a nominal tensile stress area of 0.75 times the gross area. The tensile stress area shall be calculated in accordance with the applicable ASTM standard.
[c] For end-loaded connections with a fastener pattern length greater than 38 in. (950 mm), F_{nv} shall be reduced to 83.3% of the tabulated values. Fastener pattern length is the maximum distance parallel to the line of force between the centerline of the bolts connecting two parts with one faying surface.
[d] For ASTM A307 bolts, the tabulated values shall be reduced by 1% for each 1/16 in. (2 mm) over five diameters of length in the grip.
[e] Threads assumed and permitted in shear planes in all cases.
[f] The transition area of Group 200 bolts is considered part of the threaded section.

TABLE J3.3
Nominal Hole Dimensions, in.

Bolt Diameter	Hole Dimensions			
	Standard (Dia.)	Oversize (Dia.)	Short-Slot (Width × Length)	Long-Slot (Width × Length)
1/2	9/16	5/8	9/16 × 11/16	9/16 × 1 1/4
5/8	11/16	13/16	11/16 × 7/8	11/16 × 1 9/16
3/4	13/16	15/16	13/16 × 1	13/16 × 1 7/8
7/8	15/16	1 1/16	15/16 × 1 1/8	15/16 × 2 3/16
1	1 1/8	1 1/4	1 1/8 × 1 5/16	1 1/8 × 2 1/2
≥1 1/8	d + 1/8	d + 5/16	(d + 1/8) × (d + 3/8)	(d + 1/8) × 2.5d

TABLE J3.3M
Nominal Hole Dimensions, mm

Bolt Diameter	Hole Dimensions			
	Standard (Dia.)	Oversize (Dia.)	Short-Slot (Width × Length)	Long-Slot (Width × Length)
M12	14	16	14 × 18	14 × 30
M16	18	20	18 × 22	18 × 40
M20	22	24	22 × 26	22 × 50
M22	24	28	24 × 30	24 × 55
M24	27[a]	30	27 × 32	27 × 60
M27	30	35	30 × 37	30 × 67
M30	33	38	33 × 40	33 × 75
≥M36	$d + 3$	$d + 8$	$(d + 3) \times (d + 10)$	$(d + 3) \times 2.5d$

[a]Clearance provided allows the use of a 1-in.-diameter bolt.

(d) Oversized holes are permitted in any or all plies of slip-critical connections, but they shall not be used in bearing-type connections.

(e) Short-slotted holes are permitted in any or all plies of slip-critical or bearing-type connections. The slots are permitted without regard to direction of loading in slip-critical connections, but the length shall be normal to the direction of the loading in bearing-type connections.

(f) Long-slotted holes are permitted in only one of the connected parts of either a slip-critical or bearing-type connection at an individual faying surface. Long-slotted holes are permitted without regard to direction of loading in slip-critical connections, but shall be normal to the direction of loading in bearing-type connections.

(g) Washers shall be provided in accordance with the RCSC *Specification* Section 6, except for Group 200 bolting assemblies where washers shall be provided in accordance with the applicable ASTM standard.

User Note: When Group 200 heavy-hex bolting assemblies are used, a single washer is used under the bolt head and a single washer is used under the nut. When Group 200 twist-off bolting assemblies are used, a single washer is used under the nut. Washers are of the type specified in the ASTM standard for the bolting assembly.

4. Minimum Spacing

The distance between centers of standard, oversized, or slotted holes shall not be less than 2⅔ times the nominal diameter, d, of the fastener. However, the clear distance between bolt holes or slots shall not be less than d.

User Note: A distance between centers of standard, oversized, or slotted holes of $3d$ is preferred.

TABLE J3.4
Minimum Edge Distance[a] from Center of Standard Hole[b] to Edge of Connected Part, in.

Bolt Diameter	Minimum Edge Distance
1/2	3/4
5/8	7/8
3/4	1
7/8	1 1/8
1	1 1/4
1 1/8	1 1/2
1 1/4	1 5/8
Over 1 1/4	1 1/4d

[a] If necessary, lesser edge distances are permitted provided the applicable provisions from Sections J3.11 and J4 are satisfied, but edge distances less than one bolt diameter are not permitted without approval from the engineer of record.
[b] For oversized or slotted holes, see Table J3.5.

TABLE J3.4M
Minimum Edge Distance[a] from Center of Standard Hole[b] to Edge of Connected Part, mm

Bolt Diameter	Minimum Edge Distance
12	18
16	22
20	26
22	28
24	30
27	34
30	38
36	46
Over 36	1.25d

[a] If necessary, lesser edge distances are permitted provided the applicable provisions from Sections J3.11 and J4 are satisfied, but edge distances less than one bolt diameter are not permitted without approval from the engineer of record.
[b] For oversized or slotted holes, see Table J3.5M.

5. Minimum Edge Distance

The distance from the center of a standard hole to an edge of a connected part in any direction shall not be less than either the applicable value from Table J3.4 or Table J3.4M, or as required in Section J3.11. The distance from the center of an oversized or slotted hole to an edge of a connected part shall be not less than that required for a standard hole to an edge of a connected part plus the applicable increment, C_2, from Table J3.5 or Table J3.5M.

TABLE J3.5
Values of Edge Distance Increment C_2, in.

Nominal Diameter of Fastener	Oversized Holes	Slotted Holes		
		Long Axis Perpendicular to Edge		Long Axis Parallel to Edge
		Short Slots	Long Slots[a]	
≤ 7⁄8	1⁄16	1⁄8	3⁄4d	0
1	1⁄8	1⁄8		
≥ 1 1⁄8	1⁄8	3⁄16		

[a]When the length of the slot is less than the maximum allowable (see Table J3.3), C_2 is permitted to be reduced by one-half the difference between the maximum and actual slot lengths.

TABLE J3.5M
Values of Edge Distance Increment C_2, mm

Nominal Diameter of Fastener	Oversized Holes	Slotted Holes		
		Long Axis Perpendicular to Edge		Long Axis Parallel to Edge
		Short Slots	Long Slots[a]	
≤ 22	2	3	0.75d	0
24	3	3		
≥ 27	3	5		

[a]When the length of the slot is less than the maximum allowable (see Table J3.3M), C_2 is permitted to be reduced by one-half the difference between the maximum and actual slot lengths.

User Note: The edge distances in Tables J3.4 and J3.4M are minimum edge distances based on standard fabrication practices and workmanship tolerances. The appropriate provisions of Sections J3.11 and J4 must be satisfied.

6. Maximum Spacing and Edge Distance

The maximum distance from the center of any bolt hole to the nearest edge of elements in contact shall be 12 times the thickness of the connected element under consideration, but shall not exceed 6 in. (150 mm). The longitudinal spacing of bolt holes between elements consisting of a plate and a shape, or two plates, in continuous contact shall be as follows:

(a) For painted members or unpainted members not subject to atmospheric corrosion, the spacing shall not exceed 24 times the thickness of the thinner part or 12 in. (300 mm).

(b) For unpainted members of weathering steel subject to atmospheric corrosion, the spacing shall not exceed 14 times the thickness of the thinner part or 7 in. (180 mm).

User Note: The dimensions in (a) and (b) do not apply to elements consisting of two shapes in continuous contact.

7. Tensile and Shear Strength of Bolts and Threaded Parts

The design tensile or shear strength, ϕR_n, and the allowable tensile or shear strength, R_n/Ω, of a snug-tightened or pretensioned high-strength bolt or threaded part shall be determined according to the limit states of tension rupture and shear rupture as

$$R_n = F_n A_b \tag{J3-1}$$

$$\phi = 0.75 \text{ (LRFD)} \qquad \Omega = 2.00 \text{ (ASD)}$$

where

A_b = nominal unthreaded body area of bolt or threaded part, in.^{2} (mm^2)
F_n = nominal tensile stress, F_{nt}, or shear stress, F_{nv}, from Table J3.2, ksi (MPa)

The required tensile strength shall include any tension resulting from prying action produced by deformation of the connected parts.

User Note: The available strength of a bolt in shear depends on whether the bolt is sheared through its shank or through the threads or thread runout. Bolts that are relatively short may be produced as fully threaded, without a shank, and thus may not be able to be installed in the "threads excluded" condition.

User Note: The force that can be resisted by a snug-tightened or pretensioned high-strength bolt or threaded part may be limited by the bearing or tearout strength at the bolt hole per Section J3.11. The effective strength of an individual fastener may be taken as the lesser of the fastener shear strength per Section J3.7 or the bearing or tearout strength at the bolt hole per Section J3.11. The strength of the bolt group is taken as the sum of the effective strengths of the individual fasteners.

8. Combined Tension and Shear in Bearing-Type Connections

The available tensile strength of a bolt subjected to combined tension and shear shall be determined according to the limit states of tension and shear rupture as

$$R_n = F'_{nt} A_b \tag{J3-2}$$

$$\phi = 0.75 \text{ (LRFD)} \qquad \Omega = 2.00 \text{ (ASD)}$$

where

F'_{nt} = nominal tensile stress modified to include the effects of shear stress, ksi (MPa)

$$= 1.3F_{nt} - \frac{F_{nt}}{\phi F_{nv}} f_{rv} \le F_{nt} \text{ (LRFD)} \tag{J3-3a}$$

$$= 1.3F_{nt} - \frac{\Omega F_{nt}}{F_{nv}} f_{rv} \le F_{nt} \text{ (ASD)} \tag{J3-3b}$$

F_{nt} = nominal tensile stress from Table J3.2, ksi (MPa)
F_{nv} = nominal shear stress from Table J3.2, ksi (MPa)
f_{rv} = required shear stress using LRFD or ASD load combinations, ksi (MPa)

The available shear stress of the fastener shall equal or exceed the required shear stress, f_{rv}.

User Note: Note that when the required stress, f, in either shear or tension, is less than or equal to 30% of the corresponding available stress, the effects of combined stress need not be investigated. Also note that Equations J3-3a and J3-3b can be rewritten so as to find a nominal shear stress, F'_{nv}, as a function of the required tensile stress, f_t.

9. High-Strength Bolts in Slip-Critical Connections

Slip-critical connections shall be designed to prevent slip and for the limit states of bearing-type connections. When slip-critical bolts pass through fillers, all surfaces subject to slip shall be prepared to achieve design slip resistance.

The single bolt available slip resistance for the limit state of slip shall be determined as follows:

$$R_n = \mu D_u h_f T_b n_s \tag{J3-4}$$

(a) For standard size and short-slotted holes perpendicular to the direction of the load

$$\phi = 1.00 \text{ (LRFD)} \qquad \Omega = 1.50 \text{ (ASD)}$$

(b) For oversized and short-slotted holes parallel to the direction of the load

$$\phi = 0.85 \text{ (LRFD)} \qquad \Omega = 1.76 \text{ (ASD)}$$

(c) For long-slotted holes

$$\phi = 0.70 \text{ (LRFD)} \qquad \Omega = 2.14 \text{ (ASD)}$$

where

D_u = 1.13, a multiplier that reflects the ratio of the mean installed bolt pretension to the specified minimum bolt pretension. The use of other values are permitted if approved by the engineer of record.

T_b = minimum fastener pretension given in Table J3.1 or Table J3.1M, kips (kN)

h_f = factor for fillers, determined as follows:

(1) For one filler between connected parts

$$h_f = 1.0$$

(2) For two or more fillers between connected parts

$$h_f = 0.85$$

n_s = number of slip planes required to permit the connection to slip

μ = mean slip coefficient for Class A or B surfaces, as applicable, and determined as follows, or as established by tests:

(1) For Class A surfaces (unpainted clean mill scale steel surfaces or surfaces with Class A coatings on blast-cleaned steel or hot-dipped galvanized steel whether as-galvanized or hand roughened)

$$\mu = 0.30$$

(2) For Class B surfaces (unpainted blast-cleaned steel surfaces or surfaces with Class B coatings on blast-cleaned steel)

$$\mu = 0.50$$

10. Combined Tension and Shear in Slip-Critical Connections

When a slip-critical connection is subjected to an applied tension that reduces the net clamping force, the available slip resistance per bolt from Section J3.9 shall be multiplied by the factor, k_{sc}, determined as follows:

$$k_{sc} = 1 - \frac{T_u}{D_u T_b n_b} \geq 0 \text{ (LRFD)} \tag{J3-5a}$$

$$k_{sc} = 1 - \frac{1.5T_a}{D_u T_b n_b} \geq 0 \text{ (ASD)} \tag{J3-5b}$$

where

T_a = required tension force using ASD load combinations, kips (kN)
T_u = required tension force using LRFD load combinations, kips (kN)
n_b = number of bolts carrying the applied tension

11. Bearing and Tearout Strength at Bolt Holes

The available strength, ϕR_n and R_n/Ω, at bolt holes shall be determined for the limit states of bearing and tearout, as follows:

$$\phi = 0.75 \text{ (LRFD)} \qquad \Omega = 2.00 \text{ (ASD)}$$

The nominal strength of the connected material, R_n, shall be determined in accordance with Section J3.11a or Section J3.11b.

11a. Snug-Tightened or Pretensioned High-Strength Bolted Connections

All plies of the connected elements shall be in firm contact.

(1) The nominal strength of a connected element at a bolt in a connection with standard, oversized, and short-slotted holes, independent of the direction of loading, or a long-slotted hole with the slot parallel to the direction of the bearing force, shall be the lesser of the following:

(a) Bearing

(i) When deformation at the bolt hole at service load is a design consideration

$$R_n = 2.4dtF_u \tag{J3-6a}$$

(ii) When deformation at the bolt hole at service load is not a design consideration

$$R_n = 3.0dtF_u \tag{J3-6b}$$

(b) Tearout

(i) When deformation at the bolt hole at service load is a design consideration

$$R_n = 1.2 l_c t F_u \tag{J3-6c}$$

(ii) When deformation at the bolt hole at service load is not a design consideration

$$R_n = 1.5 l_c t F_u \tag{J3-6d}$$

(2) The nominal strength of a connected element at a bolt in a connection with long-slotted holes with the slot perpendicular to the direction of force is the lesser of the following:

(a) Bearing

$$R_n = 2.0 d t F_u \tag{J3-6e}$$

(b) Tearout

$$R_n = 1.0 l_c t F_u \tag{J3-6f}$$

11b. Connections Made Using Bolts or Rods that Pass Completely Through an Unstiffened Box Member or HSS

(1) Bearing shall satisfy Section J7 and Equation J7-1

(2) Tearout

(i) For a bolt in a connection with a standard hole or a short-slotted hole with the slot perpendicular to the direction of force

(a) When deformation at the bolt hole at service load is a design consideration

$$R_n = 1.2 l_c t F_u \tag{J3-6g}$$

(b) When deformation at the bolt hole at service load is not a design consideration

$$R_n = 1.5 l_c t F_u \tag{J3-6h}$$

(ii) For a bolt in a connection with long-slotted holes with the slot perpendicular to the direction of force

$$R_n = 1.0 l_c t F_u \tag{J3-6i}$$

where

F_u = specified minimum tensile strength of the connected material, ksi (MPa)
d = nominal diameter of fastener, in. (mm)
l_c = clear distance, in the direction of the force, between the edge of the hole and the edge of the adjacent hole or edge of the material, in. (mm)
t = thickness of connected material, in. (mm)

Bearing strength and tearout strength shall be checked for both bearing-type and slip-critical connections. The use of oversized holes and short- and long-slotted holes parallel to the line of force is restricted to slip-critical connections per Section J3.3.

12. Special Fasteners

The nominal strength of special fasteners other than the bolts presented in Table J3.2 shall be verified by tests.

13. Wall Strength at Tension Fasteners

When bolts or other fasteners in tension are attached to an unstiffened box or HSS wall, the strength of the wall shall be determined by rational analysis.

J4. AFFECTED ELEMENTS OF MEMBERS AND CONNECTING ELEMENTS

This section applies to elements of members at connections and connecting elements, such as plates, gussets, angles, and brackets.

1. Strength of Elements in Tension

The design strength, ϕR_n, and the allowable strength, R_n/Ω, of affected and connecting elements loaded in tension shall be the lower value obtained according to the limit states of tensile yielding and tensile rupture.

(a) For tensile yielding of connecting elements

$$R_n = F_y A_g \tag{J4-1}$$

$$\phi = 0.90 \text{ (LRFD)} \qquad \Omega = 1.67 \text{ (ASD)}$$

(b) For tensile rupture of connecting elements

$$R_n = F_u A_e \tag{J4-2}$$

$$\phi = 0.75 \text{ (LRFD)} \qquad \Omega = 2.00 \text{ (ASD)}$$

where

A_e = effective net area as defined in Section D3, in.^{2} (mm^2)

> **User Note:** The effects of shear lag or concentrated loads dispersed within the element may cause only a portion of the area to be effective in resisting the load. For shear lag, see Chapter D.

2. Strength of Elements in Shear

The available shear strength of affected and connecting elements in shear shall be the lower value obtained according to the limit states of shear yielding and shear rupture:

(a) For shear yielding of the element

$$R_n = 0.60 F_y A_{gv} \tag{J4-3}$$

$$\phi = 1.00 \text{ (LRFD)} \qquad \Omega = 1.50 \text{ (ASD)}$$

where

A_{gv} = gross area subjected to shear, in.^{2} (mm^2)

(b) For shear rupture of the element

$$R_n = 0.60F_uA_{nv} \tag{J4-4}$$

$$\phi = 0.75 \text{ (LRFD)} \qquad \Omega = 2.00 \text{ (ASD)}$$

where

A_{nv} = net area subjected to shear, in.^{2} (mm^2)

3. Block Shear Strength

The available strength for the limit state of block shear rupture along a shear failure path or paths and a perpendicular tension failure path shall be determined as follows:

$$R_n = 0.60F_uA_{nv} + U_{bs}F_uA_{nt} \le 0.60F_yA_{gv} + U_{bs}F_uA_{nt} \tag{J4-5}$$

$$\phi = 0.75 \text{ (LRFD)} \qquad \Omega = 2.00 \text{ (ASD)}$$

where

A_{nt} = net area subjected to tension, in.^{2} (mm^2)

Where the tension stress is uniform, $U_{bs} = 1$; where the tension stress is nonuniform, $U_{bs} = 0.5$.

> **User Note:** Typical cases where U_{bs} should be taken as equal to 0.5 are illustrated in the Commentary.

4. Strength of Elements in Compression

The available strength of connecting elements in compression for the limit states of yielding and buckling shall be determined as follows:

(a) When $L_c/r \le 25$

$$P_n = F_yA_g \tag{J4-6}$$

$$\phi = 0.90 \text{ (LRFD)} \qquad \Omega = 1.67 \text{ (ASD)}$$

(b) When $L_c/r > 25$, the provisions of Chapter E apply,

where

L_c = effective length, in. (mm)
= KL
K = effective length factor
L = laterally unbraced length of the element, in. (mm)

> **User Note:** The effective length factors used in computing compressive strengths of connecting elements are specific to the end restraint provided and may not necessarily be taken as unity when the direct analysis method is employed.

5. Strength of Elements in Flexure

The available flexural strength of affected elements shall be the lower value obtained according to the limit states of flexural yielding, local buckling, flexural lateral-torsional buckling, and flexural rupture.

J5. FILLERS

1. Fillers in Welded Connections

Whenever it is necessary to use fillers in joints required to transfer applied force, the fillers and the connecting welds shall conform to the requirements of Section J5.1a or Section J5.1b, as applicable.

1a. Thin Fillers

Fillers less than ¼ in. (6 mm) thick shall not be used to transfer stress. When the thickness of the fillers is less than ¼ in. (6 mm), or when the thickness of the filler is ¼ in. (6 mm) or greater but not sufficient to transfer the applied force between the connected parts, the filler shall be kept flush with the edge of the outside connected part, and the size of the weld shall be increased over the required size by an amount equal to the thickness of the filler.

1b. Thick Fillers

When the thickness of the fillers is sufficient to transfer the applied force between the connected parts, the filler shall extend beyond the edges of the outside connected base metal. The welds joining the outside connected base metal to the filler shall be sufficient to transmit the force to the filler and the region subjected to the applied force in the filler shall be sufficient to prevent overstressing the filler. The welds joining the filler to the inside connected base metal shall be sufficient to transmit the applied force.

2. Fillers in Bolted Bearing-Type Connections

When a bolt that carries load passes through fillers that are equal to or less than ¼ in. (6 mm) thick, the shear strength shall be used without reduction. When a bolt that carries load passes through fillers that are greater than ¼ in. (6 mm) thick, one of the following requirements shall apply:

(a) The shear strength of the bolts shall be multiplied by the factor

$$1 - 0.4(t - 0.25)$$

$$1 - 0.0154(t - 6) \quad \text{(S.I.)}$$

but not less than 0.85, where t is the total thickness of the fillers.

(b) The fillers shall be welded or extended beyond the joint and bolted to uniformly distribute the total force in the connected element over the combined cross section of the connected element and the fillers.

(c) The size of the joint shall be increased to accommodate a number of bolts that is equivalent to the total number required in (b).

J6. SPLICES

Groove-welded splices in beams shall develop the nominal strength of the smaller spliced section. Other types of splices in cross sections of beams shall develop the strength required by the forces at the point of the splice.

J7. BEARING STRENGTH

The design bearing strength, ϕR_n, and the allowable bearing strength, R_n/Ω, of surfaces in contact shall be determined for the limit state of bearing (local compressive yielding) as follows:

$$\phi = 0.75 \text{ (LRFD)} \qquad \Omega = 2.00 \text{ (ASD)}$$

The nominal bearing strength, R_n, shall be determined as follows:

(a) For finished surfaces, pins in reamed, drilled, or bored holes, bolts or rods that pass completely through an unstiffened box or HSS member, and ends of fitted bearing stiffeners

$$R_n = 1.8 F_y A_{pb} \tag{J7-1}$$

where
A_{pb} = projected area in bearing, in.^{2} (mm^2)
F_y = specified minimum yield stress, ksi (MPa)

(b) For expansion rollers and rockers

(1) When $d \le 25$ in. (630 mm)

$$R_n = \frac{1.2\left(F_y - 13\right) l_b d}{20} \tag{J7-2}$$

$$R_n = \frac{1.2\left(F_y - 90\right) l_b d}{20} \tag{J7-2M}$$

(2) When $d > 25$ in. (630 mm)

$$R_n = \frac{6.0\left(F_y - 13\right) l_b \sqrt{d}}{20} \tag{J7-3}$$

$$R_n = \frac{30.2\left(F_y - 90\right) l_b \sqrt{d}}{20} \tag{J7-3M}$$

where
d = diameter, in. (mm)
l_b = length of bearing, in. (mm)

J8. COLUMN BASES AND BEARING ON CONCRETE

Provisions shall be made to transfer the column loads and moments to the footings and foundations.

In the absence of code regulations, the design bearing strength, $\phi_c P_p$, and the allowable bearing strength, P_p/Ω_c, for the limit state of concrete crushing are permitted to be taken as follows:

$$\phi_c = 0.65 \text{ (LRFD)} \qquad \Omega_c = 2.31 \text{ (ASD)}$$

The nominal bearing strength, P_p, is determined as follows:

(a) On the full area of a concrete support

$$P_p = 0.85 f_c' A_1 \tag{J8-1}$$

(b) On less than the full area of a concrete support

$$P_p = 0.85 f_c' A_1 \sqrt{A_2 / A_1} \leq 1.7 f_c' A_1 \tag{J8-2}$$

where

A_1 = area of steel concentrically bearing on a concrete support, in.^{2} (mm^2)

A_2 = maximum area of the portion of the supporting surface that is geometrically similar to and concentric with the loaded area, in.^{2} (mm^2)

f_c' = specified compressive strength of concrete, ksi (MPa)

J9. ANCHOR RODS AND EMBEDMENTS

Anchor rods shall be designed to provide the required resistance to loads on the completed structure at the base of columns including the net tensile components of any bending moment resulting from load combinations stipulated in Section B2. The anchor rods shall be designed in accordance with the requirements for threaded parts in Table J3.2.

Design of anchor rods for the transfer of forces to the concrete foundation shall satisfy the requirements of ACI 318 (ACI 318M) or ACI 349 (ACI 349M).

User Note: Column bases should be designed considering bearing against concrete elements, including when columns are required to resist a horizontal force at the base plate. See AISC Design Guide 1, *Base Plate and Anchor Rod Design*, Second Edition, for column base design information.

When anchor rods are used to resist horizontal forces, hole size, anchor rod setting tolerance, and the horizontal movement of the column shall be considered in the design.

Holes and slots larger than oversized holes and slots in Table J3.3 or Table J3.3M are permitted in base plates when adequate bearing is provided for the nut by using ASTM F844 washers or plate washers to bridge the hole.

User Note: The recommended hole sizes and corresponding washer dimensions and nuts are given in the AISC *Steel Construction Manual* and ASTM F1554. ASTM F1554 anchor rods may be furnished in accordance with product specifications with a body diameter less than the nominal diameter. Load effects such as bending and elongation should be calculated based on minimum diameters permitted by the product specification. See ASTM F1554 and the table, “Applicable ASTM Specifications for Various Types of Structural Fasteners,” in the AISC *Steel Construction Manual* Part 2.

User Note: See ACI 318 (ACI 318M) for embedment design and for shear friction design. See OSHA for special erection requirements for anchor rods.

J10. FLANGES AND WEBS WITH CONCENTRATED FORCES

This section applies to single- and double-concentrated forces applied normal to the flange(s) of wide-flange sections and similar built-up shapes. A single-concentrated force is either tensile or compressive. Double-concentrated forces are one tensile and one compressive and form a couple on the same side of the loaded member.

When the required strength exceeds the available strength as determined for the limit states listed in this section, stiffeners and/or doublers shall be provided and shall be sized for the difference between the required strength and the available strength for the applicable limit state. Stiffeners shall also meet the design requirements in Section J10.8. Doublers shall also meet the design requirements in Section J10.9.

User Note: See Appendix 6, Section 6.3, for requirements for the ends of cantilever members.

Stiffeners are required at unframed ends of beams in accordance with the requirements of Section J10.7.

User Note: Design guidance for members other than wide-flange sections and similar built-up shapes, including HSS members, can be found in the Commentary.

1. Flange Local Bending

This section applies to tensile single-concentrated forces and the tensile component of double-concentrated forces.

The design strength, ϕR_n, and the allowable strength, R_n/Ω, for the limit state of flange local bending shall be determined as

$$R_n = 6.25 F_{yf} t_f^2 \tag{J10-1}$$

$$\phi = 0.90 \text{ (LRFD)} \qquad \Omega = 1.67 \text{ (ASD)}$$

where

F_{yf} = specified minimum yield stress of the flange, ksi (MPa)

t_f = thickness of the loaded flange, in. (mm)

If the length of loading across the member flange is less than $0.15b_f$, where b_f is the member flange width, Equation J10-1 need not be checked.

When the concentrated force to be resisted is applied at a distance from the member end that is less than $10t_f$, R_n shall be reduced by 50%.

When required, a pair of transverse stiffeners shall be provided.

2. Web Local Yielding

This section applies to single-concentrated forces and both components of double-concentrated forces.

The available strength for the limit state of web local yielding shall be determined as follows:

$$\phi = 1.00 \text{ (LRFD)} \qquad \Omega = 1.50 \text{ (ASD)}$$

The nominal strength, R_n, shall be determined as follows:

(a) When the concentrated force to be resisted is applied at a distance from the member end that is greater than the full nominal depth of the member, d

$$R_n = F_{yw} t_w \left(5k + l_b\right) \tag{J10-2}$$

(b) When the concentrated force to be resisted is applied at a distance from the member end that is less than or equal to the full nominal depth of the member, d

$$R_n = F_{yw} t_w \left(2.5k + l_b\right) \tag{J10-3}$$

where

F_{yw} = specified minimum yield stress of the web material, ksi (MPa)
k = distance from outer face of the flange to the web toe of the fillet, in. (mm)
l_b = length of bearing, in. (mm)
t_w = thickness of web, in. (mm)

When required, a pair of transverse stiffeners or a doubler plate shall be provided.

3. Web Local Crippling

This section applies to compressive single-concentrated forces or the compressive component of double-concentrated forces.

The available strength for the limit state of web local crippling shall be determined as follows:

$$\phi = 0.75 \text{ (LRFD)} \qquad \Omega = 2.00 \text{ (ASD)}$$

The nominal strength, R_n, shall be determined as follows:

(a) When the concentrated compressive force to be resisted is applied at a distance from the member end that is greater than or equal to $d/2$

$$R_n = 0.80 t_w^2 \left[1 + 3\left(\frac{l_b}{d}\right)\left(\frac{t_w}{t_f}\right)^{1.5}\right]\sqrt{\frac{E F_{yw} t_f}{t_w}}\, Q_f \tag{J10-4}$$

(b) When the concentrated compressive force to be resisted is applied at a distance from the member end that is less than $d/2$

(1) For $l_b/d \le 0.2$

$$R_n = 0.40 t_w^2 \left[1 + 3\left(\frac{l_b}{d}\right)\left(\frac{t_w}{t_f}\right)^{1.5}\right]\sqrt{\frac{E F_{yw} t_f}{t_w}}\, Q_f \tag{J10-5a}$$

(2) For $l_b/d > 0.2$

$$R_n = 0.40 t_w^{\,2}\left[1+\left(\frac{4l_b}{d}-0.2\right)\left(\frac{t_w}{t_f}\right)^{1.5}\right]\sqrt{\frac{EF_{yw}t_f}{t_w}}\,Q_f \tag{J10-5b}$$

where

Q_f = chord-stress interaction parameter = 1.0 for wide-flange sections, channels, box sections, and for HSS (connecting surface) in tension
= as given in Section K1.3 for all other HSS conditions

d = full nominal depth of the member, in. (mm)

When required, a transverse stiffener, a pair of transverse stiffeners, or a doubler plate extending at least three-quarters of the depth of the web shall be provided.

4. Web Sidesway Buckling

This section applies only to compressive single-concentrated forces applied to members where relative lateral movement between the loaded compression flange and the tension flange is not restrained at the point of application of the concentrated force.

The available strength of the web for the limit state of sidesway buckling shall be determined as follows:

$$\phi = 0.85 \text{ (LRFD)} \qquad \Omega = 1.76 \text{ (ASD)}$$

The nominal strength, R_n, shall be determined as follows:

(a) If the compression flange is restrained against rotation

(1) When $(h/t_w)/(L_b/b_f) \le 2.3$

$$R_n = \frac{C_r t_w^3 t_f}{h^2}\left[1+0.4\left(\frac{h/t_w}{L_b/b_f}\right)^3\right] \tag{J10-6}$$

(2) When $(h/t_w)/(L_b/b_f) > 2.3$, the limit state of web sidesway buckling does not apply.

When the required strength of the web exceeds the available strength, local lateral bracing shall be provided at the tension flange or either a pair of transverse stiffeners or a doubler plate shall be provided.

(b) If the compression flange is not restrained against rotation

(1) When $(h/t_w)/(L_b/b_f) \le 1.7$

$$R_n = \frac{C_r t_w^3 t_f}{h^2}\left[0.4\left(\frac{h/t_w}{L_b/b_f}\right)^3\right] \tag{J10-7}$$

(2) When $(h/t_w)/(L_b/b_f) > 1.7$, the limit state of web sidesway buckling does not apply.

When the required strength of the web exceeds the available strength, local lateral bracing shall be provided at both flanges at the point of application of the concentrated forces.

In Equations J10-6 and J10-7, the following definitions apply:

C_r = 960,000 ksi (6.6×10^6 MPa), when $\alpha_s M_r < M_y$ at the location of the force
= 480,000 ksi (3.3×10^6 MPa), when $\alpha_s M_r \geq M_y$ at the location of the force
L_b = largest laterally unbraced length along either flange at the point of load, in. (mm)
M_r = required flexural strength using LRFD or ASD load combinations, kip-in. (N-mm)
M_y = yield moment about the axis of bending, ksi (MPa)
= $F_y S_x$
b_f = width of flange, in. (mm)
h = clear distance between flanges less the fillet or corner radius for rolled shapes; distance between adjacent lines of fasteners or the clear distance between flanges when welds are used for built-up shapes, in. (mm)
α_s = 1.0 (LRFD); 1.5 (ASD)

User Note: For determination of adequate restraint, refer to Appendix 6.

5. Web Compression Buckling

This section applies to a pair of compressive single-concentrated forces or the compressive components in a pair of double-concentrated forces, applied at both flanges of a member at the same location.

The available strength for the limit state of web compression buckling shall be determined as follows:

$$R_n = \left(\frac{24t_w^3\sqrt{EF_{yw}}}{h}\right)Q_f \tag{J10-8}$$

$$\phi = 0.90 \text{ (LRFD)} \qquad \Omega = 1.67 \text{ (ASD)}$$

where

Q_f = 1.0 for wide-flange sections, channels, box sections, and for HSS (connecting surface) in tension
= as given in Section K1.3 for all other HSS conditions

When the pair of concentrated compressive forces to be resisted is applied at a distance from the member end that is less than $d/2$, R_n shall be reduced by 50%.

When required, a single transverse stiffener, a pair of transverse stiffeners, or a doubler plate extending the full depth of the web shall be provided.

6. Web Panel-Zone Shear

This section applies to double-concentrated forces applied to one or both flanges of a member at the same location.

The available strength of the web panel-zone for the limit state of shear yielding shall be determined as follows:

$$\phi = 0.90 \text{ (LRFD)} \qquad \Omega = 1.67 \text{ (ASD)}$$

The nominal strength, R_n, shall be determined as follows:

(a) When the effect of inelastic panel-zone deformation on frame stability is not accounted for in the analysis

(1) For $\alpha P_r \le 0.4P_y$

$$R_n = 0.60 F_y d_c t_w \tag{J10-9}$$

(2) For $\alpha P_r > 0.4P_y$

$$R_n = 0.60 F_y d_c t_w \left(1.4 - \frac{\alpha P_r}{P_y}\right) \tag{J10-10}$$

(b) When the effect of inelastic panel-zone deformation on frame stability is accounted for in the analysis

(1) For $\alpha P_r \le 0.75P_y$

$$R_n = 0.60 F_y d_c t_w \left(1 + \frac{3 b_{cf} t_{cf}^2}{d_b d_c t_w}\right) \tag{J10-11}$$

(2) For $\alpha P_r > 0.75P_y$

$$R_n = 0.60 F_y d_c t_w \left(1 + \frac{3 b_{cf} t_{cf}^2}{d_b d_c t_w}\right)\left(1.9 - \frac{1.2\alpha P_r}{P_y}\right) \tag{J10-12}$$

where

F_y = specified minimum yield stress of the column web, ksi (MPa)
P_r = required axial strength using LRFD or ASD load combinations, kips (N)
P_y = axial yield strength of the column, kips (N)
= $F_y A_g$
A_g = gross area of member, in.^{2} (mm^2)
b_{cf} = width of column flange, in. (mm)
d_b = depth of beam, in. (mm)
d_c = depth of column, in. (mm)
t_{cf} = thickness of column flange, in. (mm)
t_w = thickness of column web, in. (mm)
α = 1.0 (LRFD); = 1.6 (ASD)

When required, doubler plate(s) or a pair of diagonal stiffeners shall be provided within the boundaries of the rigid connection whose webs lie in a common plane.

See Section J10.9 for doubler plate design requirements.

7. Unframed Ends of Beams and Girders

At unframed ends of beams and girders not otherwise restrained against rotation about their longitudinal axes, a pair of transverse stiffeners, extending the full depth of the web, shall be provided.

8. Additional Stiffener Requirements for Concentrated Forces

Stiffeners required to resist tensile concentrated forces shall be designed in accordance with the requirements of Section J4.1 and welded to the loaded flange and the web. The welds to the flange shall be sized for the difference between the required strength and available strength. The stiffener to web welds shall be sized to transfer to the web the algebraic difference in tensile force at the ends of the stiffener.

Stiffeners required to resist compressive concentrated forces shall be designed in accordance with the requirements in Section J4.4 and shall either bear on or be welded to the loaded flange and welded to the web. The welds to the flange shall be sized for the difference between the required strength and the applicable limit state strength. The weld to the web shall be sized to transfer to the web the algebraic difference in compression force at the ends of the stiffener. For fitted bearing stiffeners, see Section J7.

Transverse full depth bearing stiffeners for compressive forces applied to a beam flange(s) shall be designed as axially compressed members (columns) in accordance with the requirements of Section E6.2 and Section J4.4. The member properties shall be determined using an effective length of $0.75h$ and a cross section composed of two stiffeners and a strip of the web having a width of $25t_w$ at interior stiffeners and $12t_w$ at the ends of members. The weld connecting full depth bearing stiffeners to the web shall be sized to transmit the difference in compressive force at each of the stiffeners to the web.

Transverse and diagonal stiffeners shall comply with the following additional requirements:

(a) The width of each stiffener plus one-half the thickness of the column web shall not be less than one-third of the flange or moment connection plate width delivering the concentrated force.

(b) The thickness of a stiffener shall not be less than one-half the thickness of the flange or moment connection plate delivering the concentrated load nor less than the width divided by 16.

(c) Transverse stiffeners shall extend a minimum of one-half the depth of the member except as required in Sections J10.3, J10.5, and J10.7.

9. Additional Doubler Plate Requirements for Concentrated Forces

Doubler plates required for compression strength shall be designed in accordance with the requirements of Chapter E.

Doubler plates required for tensile strength shall be designed in accordance with the requirements of Chapter D.

Doubler plates required for shear strength (see Section J10.6) shall be designed in accordance with the provisions of Chapter G.

Doubler plates shall comply with the following additional requirements:

(a) The thickness and extent of the doubler plate shall provide the additional material necessary to equal or exceed the strength requirements.

(b) The doubler plate shall be welded to develop the proportion of the total force transmitted to the doubler plate.

10. Transverse Forces on Plate Elements

When a force is applied transverse to the plane of a plate element, the nominal strength shall consider the limit states of shear and flexure in accordance with Sections J4.2 and J4.5.

User Note: The flexural strength can be checked based on yield-line theory and the shear strength can be determined based on a punching shear model. See AISC *Steel Construction Manual* Part 9 for further discussion.